

17 A ball is thrown upwards and reaches a maximum height.
The ball then falls and bounces repeatedly.

After the n th bounce, the ball reaches a height of h_n
After the next bounce, the ball reaches a height given by $h_{n+1} = 0.55h_n$

After the 1st bounce, the ball reaches a height of 8 metres.

What height does the ball reach after the 4th bounce?

..... metres